

THE KEY TO RESIDENTIAL ENERGY EFFICIENCY

Most of you reading this magazine are probably lucky enough to have energy conservation efforts in your area. Many of these efforts take the form of programs that encourage, support, and incentivize energy saving but do not actually install energy saving measures. Some of these programs are state sponsored, many operate through the various utilities, and some are even regionally or nationally based. No matter who funds or designs the programs, the goals are almost always the same: to lower the consumer's energy bill, reduce total load growth, mitigate environmental damage, and reduce dependence on scarce fossil fuels. But in our experience in the weatherization and energy services fields, the divide between conservation programs and the homeowner is often too vast, making energy conservation goals much more difficult to achieve than they need to be.

Bridging a Divide

There is one group that comes between the program designers and the average homeowner. That group is weatherization and HVAC contractors. These people walk between the two worlds. They know that saving energy should have a higher priority than it does, and they also know that homeowners generally would rather buy a new TV than install attic insulation or have their ducts sealed. Contractors know what needs to be done to save energy, and their livelihoods depend on doing so; this gives them the motivation and focus that homeowners often lack.

Perhaps, then, the intermediate goal of an energy conservation program should be to make these contractors successful. The happy byproduct of a more successful contractor will be lower energy bills, reduced total-load

growth, less environmental damage, and reduced dependence on scarce fossil fuels. For the program, this focus on an intermediate goal will mean reaching a higher end goal with less effort. After all, it is far easier to help a handful of even the most stubborn contractors to succeed than it is to convince millions of homeowners to wear a sweater rather than turn up the thermostat. Stated differently, it is far easier to help an interested contractor make money than it is to encourage an uninterested homeowner to spend money.

What can programs do to help contractors be successful? The first step is for both program designers and contractors to realize that they are all on the same team, and that they all sink or swim together. How often do you define these groups as "us" and "them"? Does the following sound familiar? Contractors: "The programs have too much paperwork, too many rules with too many changes, don't give high enough incentives, and don't communicate well with contractors." Program administrators and designers: "Contractors don't fill out the paperwork right, they don't listen, contractors always want more money, and you don't get quality feedback from contractors." Does any of that sound familiar?

Program designers and contractors should strive to treat each other with mutual respect. They might even become friends. This can be hard. Even though contractors and program designers have some insight into each other's worlds, they still look at things in fundamentally different ways, speaking different languages, and responding to different pressures.

But out of a positive relationship, program designers and local contractors can work together to create a system of training, incentives, marketing, and quality control that helps the homeowner, the contractor, and the



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Abel Sanchez of Total Comfort Weatherization in Salem, Oregon, installs a new, energy-efficient window. Under the window are freshly patched holes that were used for injecting insulation.

program by encouraging more high quality installations of conservation measures.

The Conservation Market

What is the real common goal of program designers and HVAC and weatherization contractors? Contractors want to sell more jobs with less hassle. Program people want energy savings and market transformation. When you translate these goals into a common language, you get something like "We want to invigorate the marketplace for energy conservation."



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Carlos Wendlandt of Total Comfort prepares to inject insulation into the walls of an existing home.

How do you do this? A market is defined as a group of people getting together in a place for the purpose of mutually beneficial trade. Breaking that down a bit, a market requires three things to be vigorous: people, a place, and mutually beneficial trade. We wrote about the people above. A good relationship between the program developers, contractors, and homeowners is key.

Markets happen in places, but “place” can mean different things to different people. This leads to competing pressures. On the one hand, conservation markets are different in Arizona than they are in Alaska; they are even different in the mountains than in the valleys 50 miles away. So it makes sense to have local control over programs, right? But what about the contractors who work in multiple markets? Don’t they need some uniformity? What if you are a national builder, or even just an independent contractor who operates on the edge of a utility service district, or even on the border between two states? What is the answer? You must reach a happy medium between local control and uniformity—and that will be much easier to do working with your new friends.

Even if you get people together in a place, there won’t be a market until

there is mutually beneficial trade. One of two things prevents real trade. Most commonly, both homeowners and program designers see energy saving measures as too expensive, given the cost of the energy saved, except in a limited set of circumstances. The other problem can happen even when measures are installed and incentives are given. That is when an incentive is given on a project that would have been done without that incentive. In this case the homeowner is getting a benefit that played no part in the decision to do the project and the program is recording savings that would have happened without its efforts. In other words, both the homeowner and program are getting a free ride. Between savings being too expensive for either group, and savings being too cheap for homeowners, there isn’t a very big window for programs to have a real effect.

Market Transformation

A more vigorous market and true program efficacy comes into being when the program and the homeowner are brought together by the weatherization and HVAC contractors. These contractors can trade energy savings to program designers in exchange for end user incentives, cooperative marketing dollars, promotions, training assistance, and so forth. Contractors can also trade their services to homeowners for money (that is their job). Strange as it seems, this three-way trade can gain more savings for the program, and thereby better achieve its end goals.

Why does this tripartite market succeed? The reason is simple. It is the same reason why programs have such a hard time convincing homeowners to act. The jump from one large program to thousands or millions of individual homeowners is too great. It is very difficult for any program to give enough attention to a specific homeowner to show that homeowner why energy conservation pays. Weatherization and HVAC

contractors, on the other hand, are masters at showing the value of specific conservation measures to specific people, and at helping them to act in a specific time frame.

Programs with nothing more than broad goals and broad audiences will never achieve their potential. The successful program will focus its efforts on helping weatherization and HVAC contractors to succeed. What this means will vary with location and with the specific people involved. But the key to finding these solutions will be the same everywhere. Program designers and contractors need to work together. Don’t just copy what worked somewhere else; each market offers unique opportunities. The best way to capitalize on these opportunities is to involve the contractors in the program design. Whenever there is a brainstorming session or a decision-making meeting about program design, there should be a group of contractors who are asked, “How can this program best help you succeed?”

Program planners need to keep in mind that every program decision will affect the livelihood of the contractors who do the work of the program. One rule change could put someone out of business, or it could lead to 50 new jobs. Contractors should remember that program designers are trying to help them by subsidizing their business. Help them and thank them. Each group is doing a favor for the other, and each group is benefiting. This is a market, and by working together more closely, program planners and contractors can make it a more successful one.



—Jeremy Anderson and
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